

Stress, Hormone, and Energy

Stress $\rightarrow$ HPA Axis

HPA

Hypothalamus

Pituitary



Linear Algebra

Chapter 1: Linear Equations

1.1 Systems of Linear Equations



Formal Languages

Introduction

1.1

1.2

1.3



$\mathbb{Q} \rightarrow \mathbb{R}$ via language

Example: $\mathbb{Q} \rightarrow \mathbb{R}$ via

$\mathbb{Q} \rightarrow \mathbb{R}$ via



Algorithm 1: Finding the minimum

Input: $A[1:n]$, min



is a language

is a language

is a language







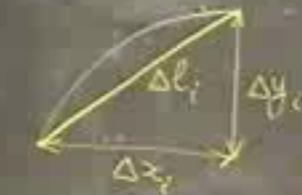
$$\Delta l_i = \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2}$$

$$= \sqrt{\frac{\Delta x}{\Delta t_i}}$$



$t = 1, \dots, N$

$$\Delta l_i = \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2}$$



$i = 1, \dots, N$

$$\Delta l_i = \sqrt{\Delta x_i^2 + \Delta y_i^2}$$



Arc Length of Parametric Curves



$$x = f(t)$$

$$y = g(t)$$

Arc Length of Parametric Curves



Arc Length





100101



for

$2, 2^2$

$$N_s = 2^n$$

$$\log_2 N_s = n$$

$$\log_2 N_s$$

bro let

$\vec{u}, \vec{v}$

$$N_S = 2^n$$

or $1 \rightarrow$

$$0 \rightarrow 1 \rightarrow N_S = n$$

$0, 1$

$00, 01, 10, 11$

$2^0, 2^1$

$|0\rangle$

or

$$N_s = 2^n$$

$|00\rangle$

N_s

a ket

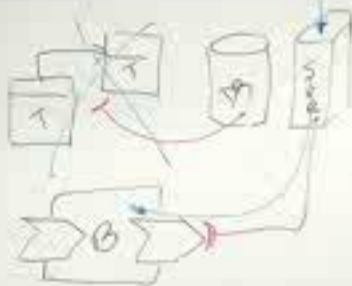
$2, 2^2$

N_s

$|1\rangle$

$|1\rangle$

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
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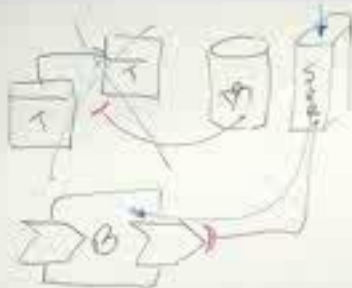


Exhaust
Volume
Pressure

Exhaust
Fuel



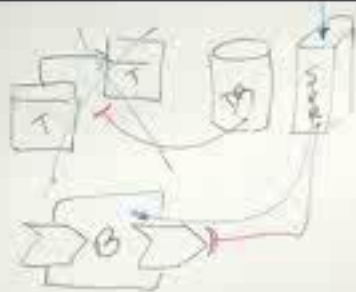
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	-----



Current
Value
Process



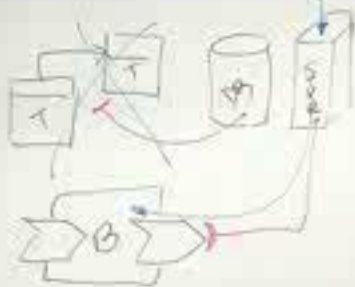
DATE	TIME	BY
10/10/17	10:10	101
10/10/17	10:10	101
10/10/17	10:10	101
10/10/17	10:10	101



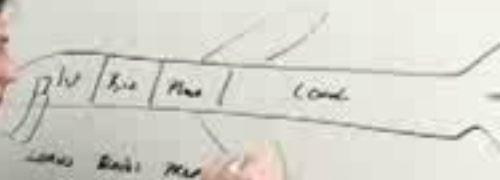
Outputs



Q. 1. 2.	
Q. 2. 3.	
Q. 3. 4.	
Q. 4. 5.	
Q. 5. 6.	
Q. 6. 7.	
Q. 7. 8.	
Q. 8. 9.	
Q. 9. 10.	



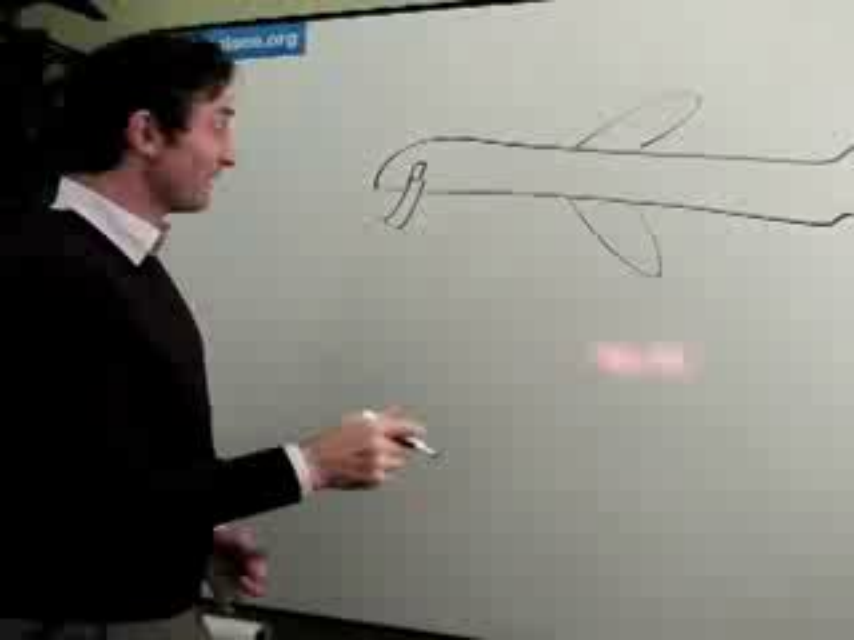






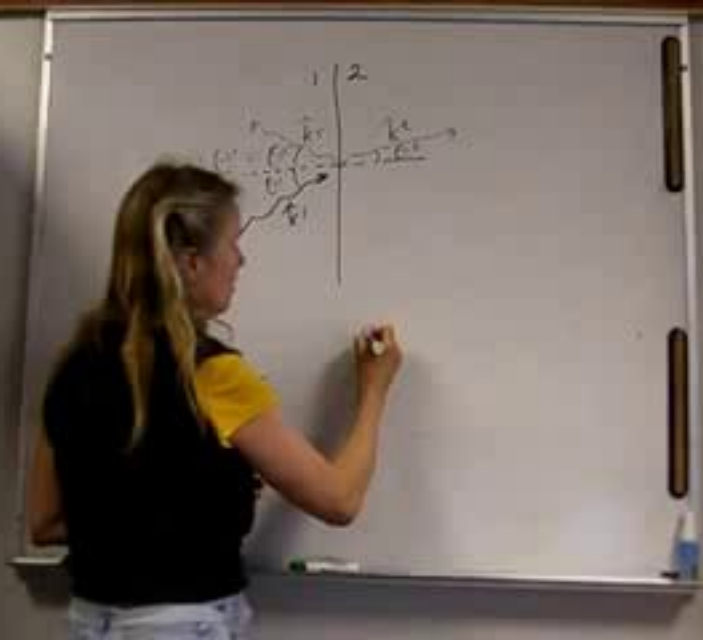














Snell's Law
 $n_i \sin \theta_i = n_r \sin \theta_r$

n_i



Snell's Law

$$c_1 = c_2$$

$$n_1 \sin \theta_i = n_2 \sin \theta_r$$



Snell's Law

$$c_1 = c_2$$

$$n_1 \sin \theta_i = n_2 \sin \theta_r$$

$$n_1 \sin \theta_i = n_2 \sin \theta_r$$



Snell's Law

$\sin \theta_1$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

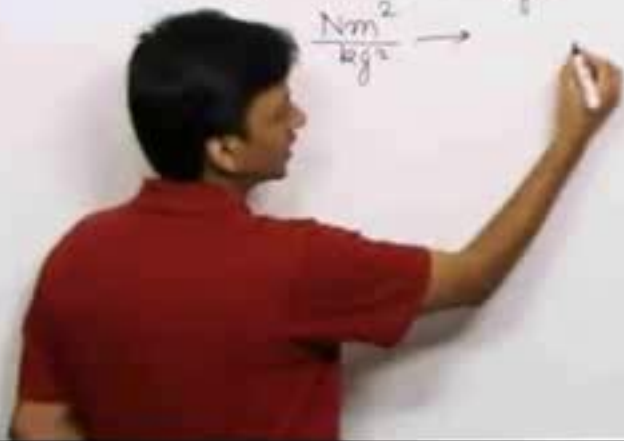
$$\sin \theta_2 = \frac{n_1}{n_2} \sin \theta_1$$

$$\theta_2 = \sin^{-1} \left(\frac{n_1}{n_2} \sin \theta_1 \right)$$

Q1

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$

$$\frac{\text{Nm}^2}{\text{kg}^2} \rightarrow$$

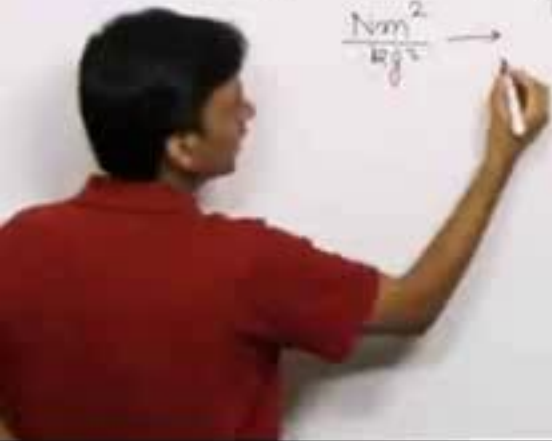


Q1

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$

$$N \rightarrow \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

$$\frac{\text{Nm}^2}{\text{kg}^2} \rightarrow$$



Q1

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$

$$N \rightarrow \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

$$\frac{\text{Nm}^2}{\text{kg}^2} \rightarrow \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \times \frac{\text{m}^2}{\text{kg}^2}$$

Q1

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2} \quad \text{N} \rightarrow \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

$$\frac{\text{Nm}^2}{\text{kg}^2} \rightarrow \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \times \frac{\text{m}^2}{\text{kg}^2} \rightarrow \frac{\text{m}^3}{\text{s}^2 \cdot \text{kg}}$$

Q1

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$

$$N \rightarrow \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

$$\frac{\text{Nm}^2}{\text{kg}^2} \rightarrow \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \times \frac{\text{m}^2}{\text{kg}^2} \rightarrow \frac{\text{m}^3}{\text{s}^2 \cdot \text{kg}}$$

$$\frac{\text{m}^3}{\text{s}^2 \cdot \text{kg}}$$

Q1

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2} \quad \text{N} \rightarrow \text{kg} \frac{\text{m}}{\text{s}^2}$$

$$\frac{\text{Nm}^2}{\text{kg}^2} \rightarrow \frac{\text{kg} \frac{\text{m}}{\text{s}^2} \times \text{m}^2}{\text{kg}^2} \rightarrow \frac{\text{m}^3}{\text{s}^2 \text{kg}}$$

$$\frac{\text{m}^3}{\text{s}^2 \text{kg}} \rightarrow \frac{\text{cm}^3}{\text{s}^2 \text{g}}$$

Q1

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2} \quad \text{N} \rightarrow \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

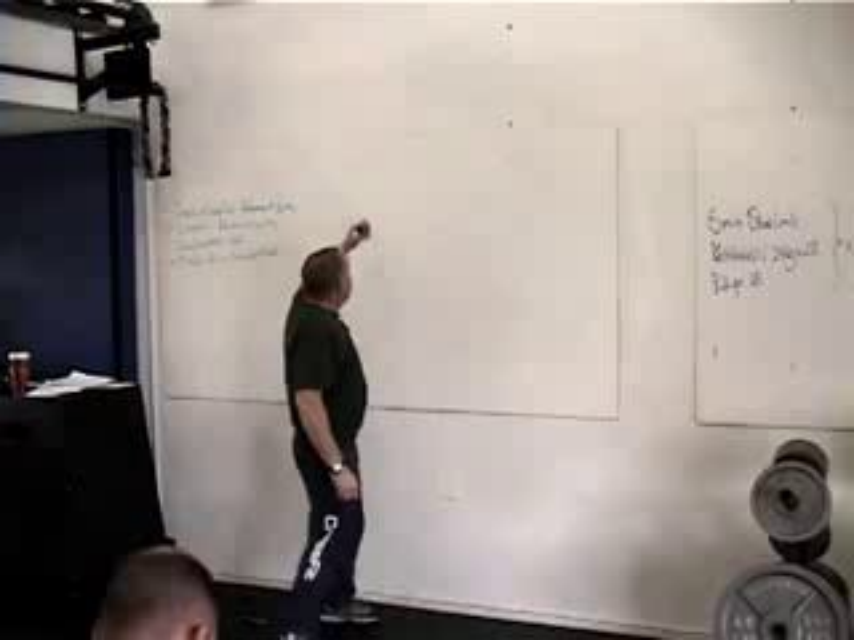
$$\frac{\text{Nm}^2}{\text{kg}^2} \rightarrow \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \times \frac{\text{m}^2}{\text{kg}^2} \rightarrow \frac{\text{m}^3}{\text{s}^2 \cdot \text{kg}}$$

$$\frac{\text{m}^3}{\text{s}^2 \cdot \text{kg}} \rightarrow \frac{\text{cm}^3}{\text{s}^2 \cdot \text{g}}$$

$$1 \text{ m} = 100 \text{ cm}$$

Handwritten notes on the left whiteboard, including the words "Handwritten", "Notes", and "Handwritten".

Handwritten notes on the right whiteboard, including the words "Handwritten", "Notes", and "Handwritten".



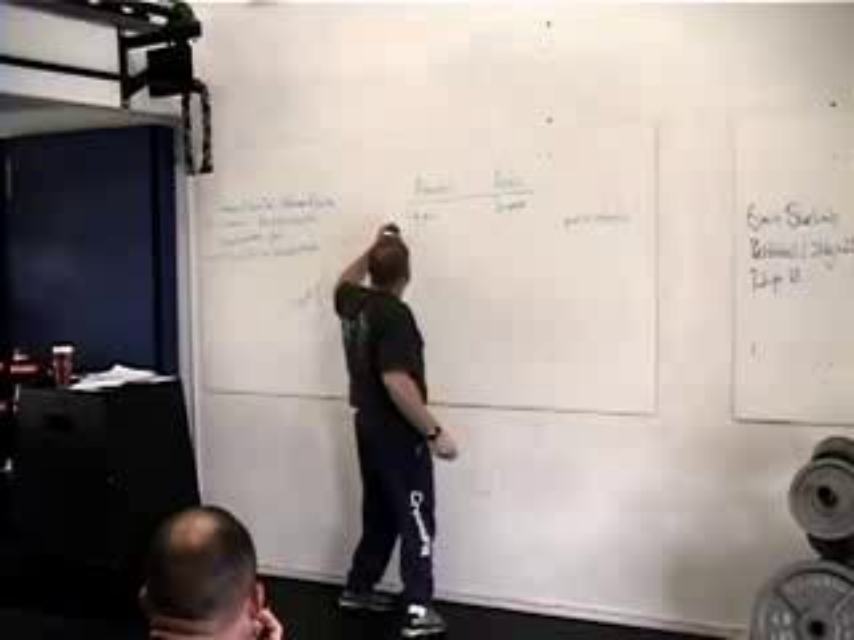


Handwritten text on the left whiteboard, possibly a list or notes.

Handwritten text in the center whiteboard section, being written by the man.

Handwritten text on the right whiteboard, including a list or notes.





Handwritten text on the left side of the whiteboard, possibly a list or notes.

Item	Price
...	...

Handwritten text on the right side of the whiteboard, possibly a list or notes.



TODAY

TAKING BETTER
LECTURE NOTES

- NEXT WEEK:

HOW TO PREPARE
YOUR QUESTIONS

Dr. [unclear]
[unclear]



TODAY

TAKING BETTER

LECTURE NOTES

- NEXT

HOW

PREDICT

THE

RESULTS

1) REVIEW
NOTES

2) PREVIEW
TEXT

TAKING BETTER
LECTURE NOTES

NOTES

REVIEW

TEXT

ADDRESS

NEXT WEEK

HOW TO PREDICT

TEST QUESTIONS



TODAY:

TAKING BETTER

LECTURE NOTES

- NEXT

HOW

NOT

TO

1) REVIEW
NOTES

2) PREVIEW
TEXT

USE ABBREV.

Reconstruction

Rec.

Rec.

(R)

TAKING BETTER

LECTURE NOTES:

WEEK:

TO PREDICT

QUESTIONS

NOTES:

2) REVIEW

TEXT

Reconstruction

Recen.

- USE ARREAR

Rec.

R	R	D
R	R	D
R	R	D

TAKING BETTER

LECTURE NOTES

- NEXT
- HIGHER
- NEXT
- NEXT

1) REVIEW

2) PREVIEW

- USE APPROPRIATE

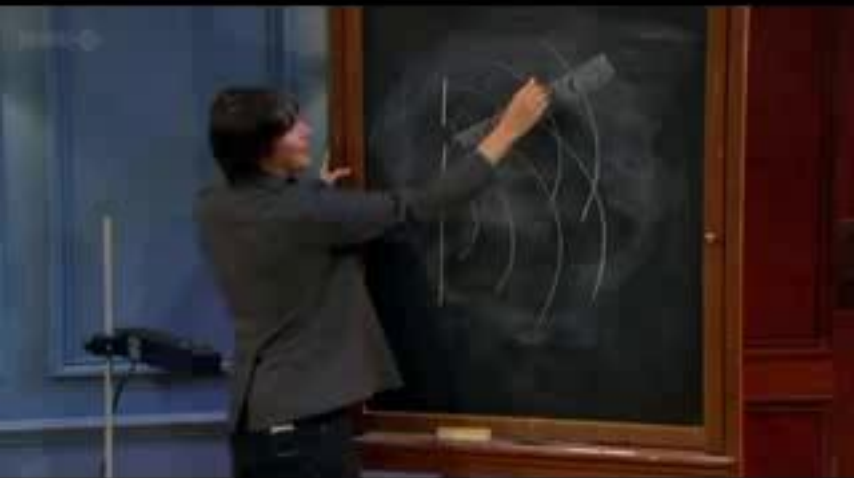
3) RECOMMENDATION

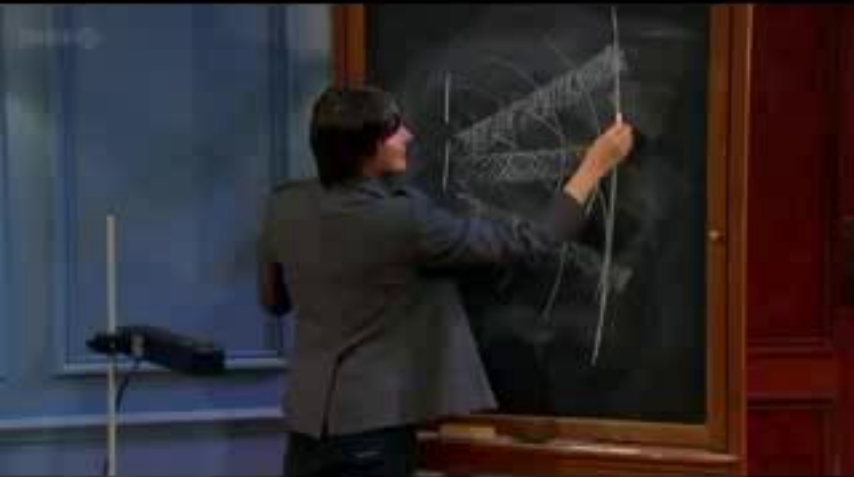
Room

Room

(R)

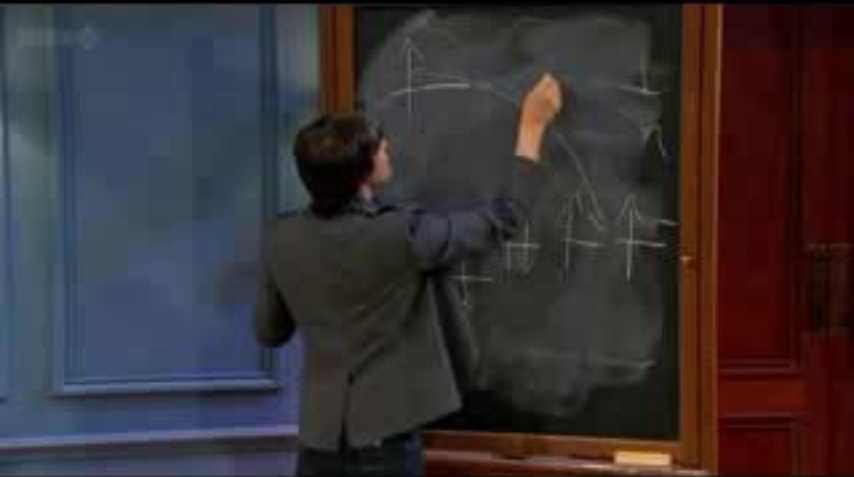
















$$v_{\text{Sound}} \approx 340 \text{ m/s}$$

approach $f' > f$

receding

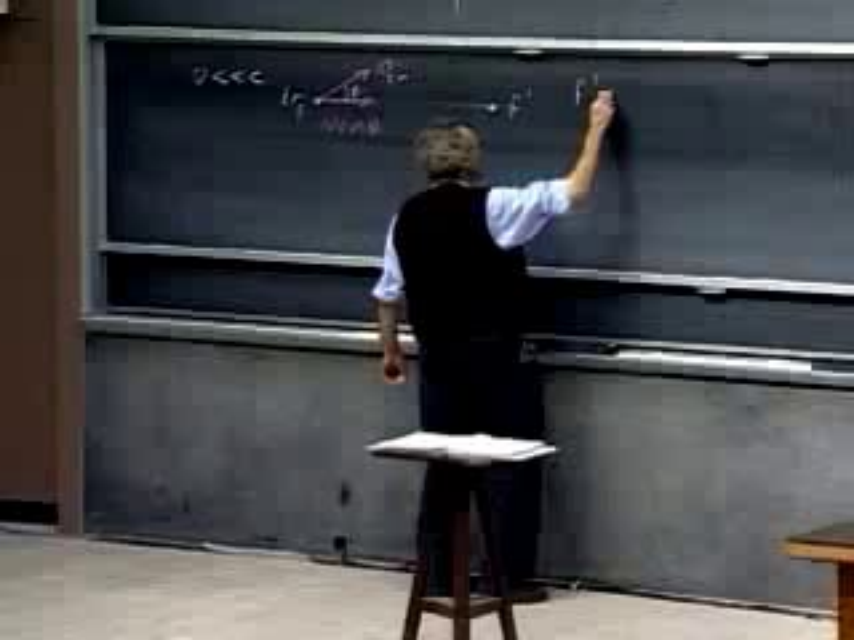
$v_{\text{sound}} = 340 \text{ m/s}$
approach $f' > f$ $f' = 1.00 f$
receding $f' < f$ $f' = 0.99 f$
 $v_s = 10 \text{ m/s}$

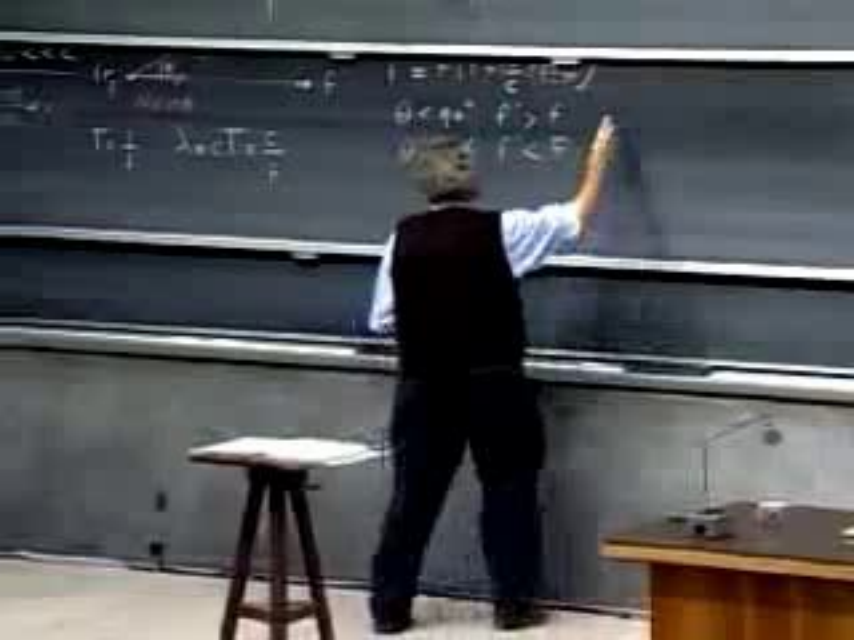




Supply Effect

U_{sound} =
approach f
reading f
U = 100 kg





$$\begin{array}{ccc} r_1 & \triangle & r_2 \\ & \Delta r_{12} & \\ & \text{---} & \\ & r_3 & \end{array}$$

$$T = \frac{1}{\lambda} \lambda c T = \frac{c}{\lambda}$$

$$f$$

$$I = \frac{P}{A} = \frac{P}{\pi r^2}$$

$$\theta < \theta_0 \Rightarrow f > f_0$$

$$\theta > \theta_0 \Rightarrow f < f_0$$









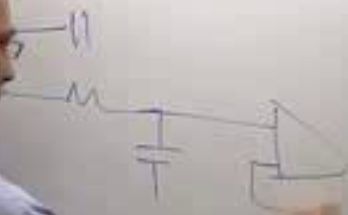
Filtering 101 - Mult: H
Sallen



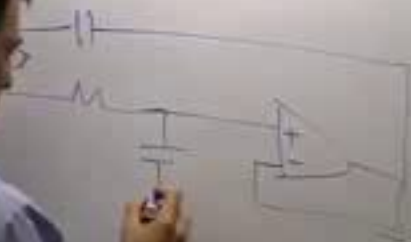
Filtering 101 - Mult: P
Sallen



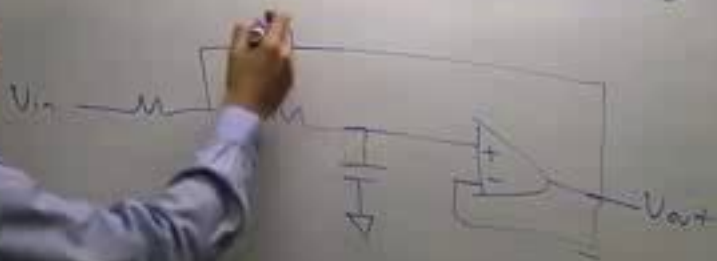
Filtering 101 - Mult: H
Sallen

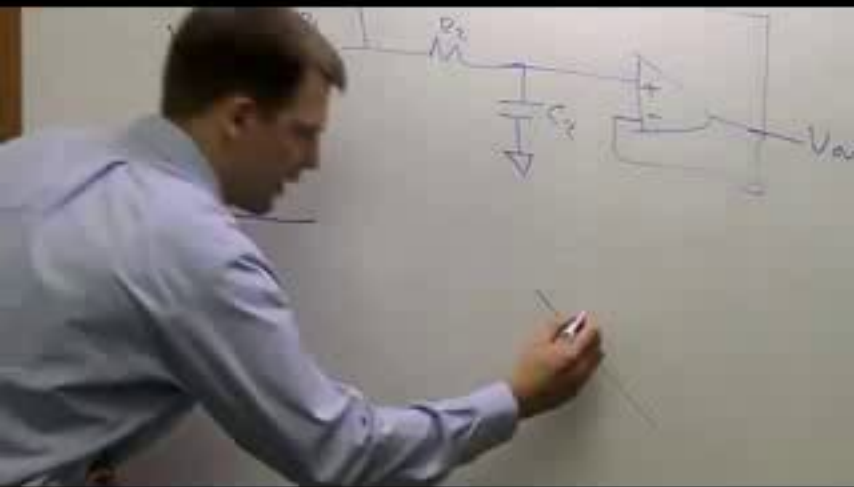


Filtering 101 - Mult. Pole
Sallen Key



Filtering 101 - Miller 101
Sallen key





robor_fc@yacc-

65-052-2811725

7/1/08

③ MULTIPLICATIONS (w/ ALL DIGITS '2')



1000 - 12 @ 1000 -

- 2811723

6 MULTIPLICATION: (2ND 1000 DIGITS 2')

$$\begin{array}{r} 51 \\ \times 22 \\ \hline \end{array}$$

10

Today - 12/5/2020

63-232-2311723

6 MULTIPLICATION: (2ND NO. (DIGITS "2")

$$\begin{array}{r} 51 \\ \times 22 \\ \hline \end{array}$$

STEPS:

① $1 \times 2 = 2$

②

③

TODAY - 12/5/2000

062 - 2311733

2nd

6. MULTIPLICATIONS/2

4.12.615 (2)

$$\begin{array}{r} 51 \\ \times 22 \\ \hline \end{array}$$

STEPS:

① $1 \times 2 = 2$

③ $51 \times 2 = 102$







→ NMS





18.01 UNIT 1

(available after

BLACKBOARD E

$\frac{1}{2} \times \$8.75$ per

lottery after



18.01 UNIT 1 DIFFERENTIATION.



18.01 UNIT 1 DIFFERENTIATION.
A. WHAT IS A DERIVATIVE?



18.01 UNIT 1 DIFFERENTIATION.

A WHAT IS A DERIVATIVE?
geom. interp.



- geometric interp.
- physical interp.



18.01 UNIT 1 DIFFERENTIATION.

A. WHAT IS A DERIVATIVE?

- geom. interp.
- physical interp.
- importance to all measurements.

(←

18.01 UNIT 1 DIFFERENTIATION.

A. WHAT IS A DERIVATIVE?

- geom. interp.
- physical interp.
- importance to all measurements.
(sci, eng, econ, polit sci)



Tag Questions



Tag Questions

Q. You are from

Tag Questions

① You are from India, aren't you?

Tag Questions

1) You ⁺are from India, aren't you?

2) You ⁺like pizza, don't you?

(2) - Guttag & Grimson
web.mit.edu/6.00.10/www

Skills: Computational thinking

①

6.00 - Guttag & Grimmer
web.mit.edu/6.00/6.00.01

①

Skills
Computational thinking
understand code
understand abilities & limits

Computation

Declarative
 $\sqrt{x}$ is y s.t.

Imperative.





Label comp - Fixed prog. comp

(3)

balanced comp. $\rightarrow$ fixed prop. comp

- calculator

- Atan

limited comp - Fixed prog. comp

- calculator
- Atanasoff (1941) - linear eqs.
- T

web page.

www

web page.

→ www.stanford.edu/class/cs106

- cs106

web page.

→ www.stanford.edu/class/cs106a

→ cs106a.stanford.edu

Sections

web page.

→ www.stanford.edu/class/cs106a

→ cs106a.stanford.edu

Sections

cs198.stanford.edu/section
5pm Thurs



length
 m
 L

time
 sec
 T

mass
 kg
 M

[speed]

A man in a blue shirt and khaki pants is standing with his back to the camera, writing on a chalkboard. He is holding a piece of chalk in his right hand and has just finished writing the word "[speed]" in white chalk. The chalkboard is dark and has some horizontal lines. Above him, there are three columns of handwritten text in white chalk, each representing a physical quantity and its units. The first column is for length, with units m and L. The second column is for time, with units sec and T. The third column is for mass, with units kg and M.

$$[speed] = \frac{[L]}{[T]}$$

$$[vol] = L$$

$$[\text{speed}] = \frac{[L]}{[T]} \quad [\text{density}]$$

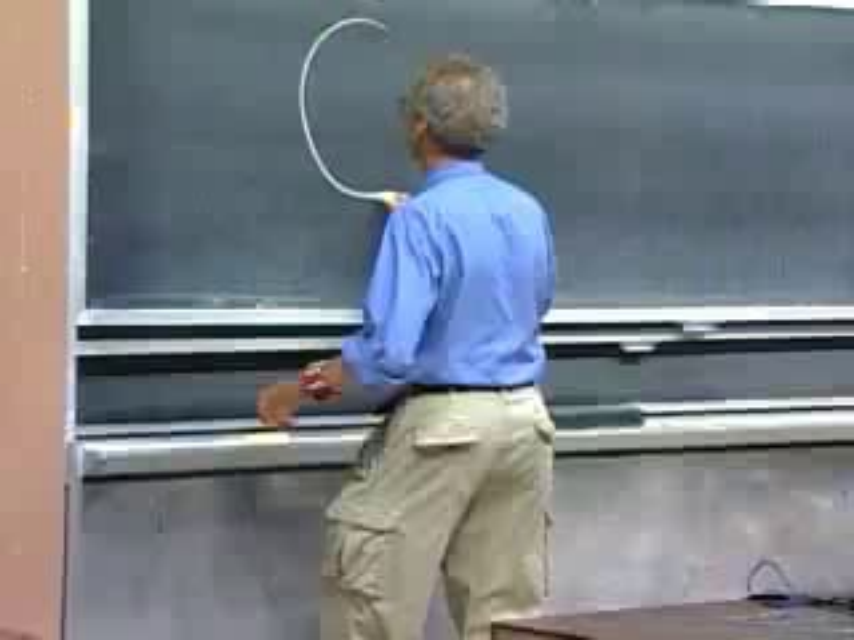
$$[\text{volume}] = [L^3]$$

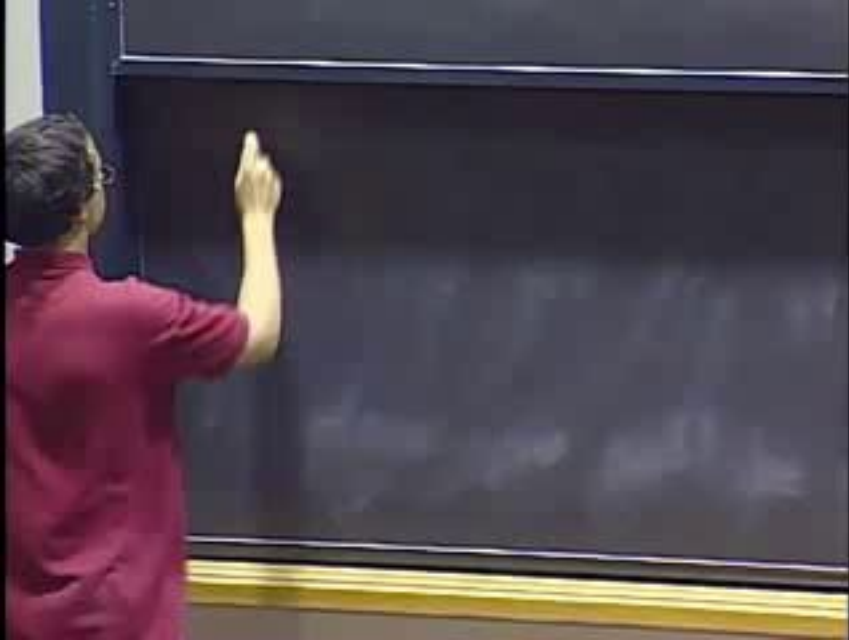
$$\begin{aligned} [\text{speed}] &= \frac{[L]}{[T]} & [\text{density}] &= \frac{[M]}{[L]^3} \\ [\text{volume}] &= [L]^3 & [\text{acc}] &= \end{aligned}$$



AI test







Vector



Vectors

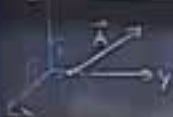


- direction

- magnitude / length

$$\vec{A} =$$

Vectors:



= direction

= magnitude length

$$\vec{A} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$$



Vectors:

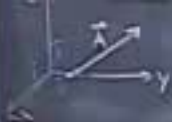


{ - direction
- magnitude / length

$$\vec{A} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k} = \langle a_1, a_2, a_3 \rangle$$

Length = $|\vec{A}|$ (a scalar) ; direction: $\text{dir}(\vec{A})$





l - magnitude / length

$$\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k} = \langle a_1, a_2, a_3 \rangle$$

Length $|\vec{A}|$ (a scalar), direction $\text{dir}(\vec{A})$



Length of $\vec{A} =$



Q Length of $\vec{A} = \langle 3, 2, 17 \rangle$?



Final Exam Dec 11, 12:30 - 2:30 pm
at Hearst Gym!

S





dim $D = 1$

ambient space: $\mathbb{R}$



dim $D = 1$

ambient space: $\mathbb{R}$

$\int_a^b$

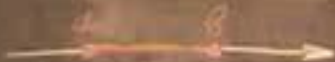
$$= f(b) - f(a)$$

dim $D = 1$

ambient space: $\mathbb{R}$


$$\int_a^b f'(x) dx = f(b) - f(a)$$

• ambient space: $\mathbb{R}$



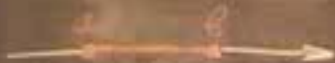
$$\int_a^b f'(x) dx = f(b) - f(a)$$

• ambient space: $\mathbb{R}^2$



=

• ambient space: $\mathbb{R}$



$$\int_a^b f'(x) dx = f(b) - f(a)$$

• ambient space: $\mathbb{R}^2$



$$= f(B) - f(A)$$

Optimistically

I

L

O

Judge

Opportunity

~



Mr E's
Listening
Lesson

Optimistically



(emotion)

not Delivery

quality

Optimistically

Ideas

Longer

Open mind ()

Judge content

Opportunisticly



Optimistically

Ideas

Longer

Open mind (emotion)

Judge content not Larry

Opportunisticly



Optimistically

Ideas

Longer

Open mind (emotion)

Judge content not Day

Opportunistically



Optimistically

Ideas

Longer

Open mind (a)

Judge content

Opportunisticly

Some
people



Ideas

Longer

Open mind (emotion)

Judge content not Delivery

Opportunistically

Some people
are not good
speakers



Listening
Lesson

1/1 E's

gebra
g
linear Alg
edu

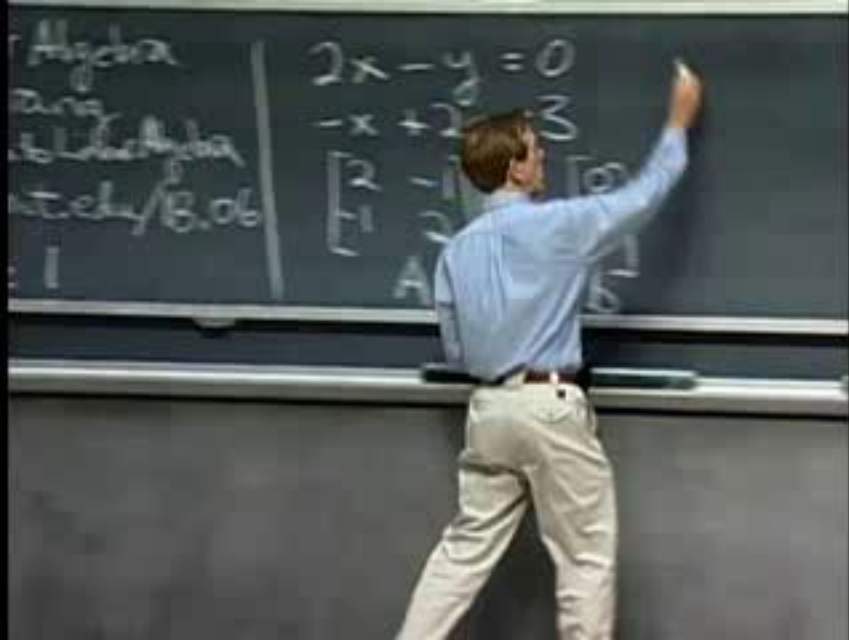


ebra

near A

l

$$2x - y = 0$$



Algebra
and
Algebra
nt.edu/B.06

$$2x - y = 0$$
$$-x + 2 = 3$$
$$\begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

A

Row picture



$$2x - y = 0$$

$$-x + 2y = 2$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

A

Row picture $2x - y = 0$



$$2x - y = 0$$

$$-x + 2y = 3$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$A \quad X = b$$

Row picture $2x - y = 0$



Linear Algebra
 + Strang
 Introduction to Linear Algebra
 web.mit.edu/18.06
 Lecture 1

$$2x - y = 0$$

$$-x + 2y = 3$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$A \mathbf{x} = \mathbf{b}$$

Row picture $2x - y = 0$



equations, n unknowns
 Lecture
 picture
 topic

$$\begin{bmatrix} 2 \\ -1 \end{bmatrix} + y$$

1911



1911 E. Rutherford



1911 E. Rutherford



See you

He





$n = 287$ $p =$



